

Parcial 15/06/2024

1)

$$T(n) = \begin{cases} 1 & n=1 \\ 25T(n/5) + n^2 & n > 1 \end{cases}$$

Paso 1

$$25T\left(\frac{n}{5}\right) + n^2$$

$$5^2 T\left(\frac{n}{5}\right) + n^2$$

Paso 2

$$5^2 \left[5^2 T\left(\frac{n}{5}\right) + \left(\frac{n}{5}\right)^2 \right] + n^2$$

$$5^2 \left[5^2 T\left(\frac{n}{5^2}\right) + \frac{n^2}{5^2} \right] + n^2$$

$$5^2 \cdot 5^2 T\left(\frac{n}{5^2}\right) + 5^2 \cdot \frac{n^2}{5^2} + n^2$$

$$5^4 T\left(\frac{n}{5^2}\right) + n^2 + n^2$$

$$5^4 T\left(\frac{n}{5^2}\right) + 2n^2$$

Paso 3

$$5^4 \left[5^2 T\left(\frac{n}{5^3}\right) + \left(\frac{n}{5^2}\right)^2 \right] + 2n^2 = 5^4 \left[5^2 T\left(\frac{n}{5^3}\right) + \frac{n^2}{5^4} \right] + 2n^2 =$$

$$= 5^4 \cdot 5^2 T\frac{n}{5^3} + 5^4 \cdot \frac{n^2}{5^4} + 2n^2 = 5^6 T\frac{n}{5^3} + n^2 + 2n^2 = 5^6 T\frac{n}{5^3} + 3n^2$$

$$\text{Final paso 3} = 5^6 T\left(\frac{n}{5^3}\right) + 3n^2$$

$$\text{Paso } i \quad 5^{i \cdot 2} T\left(\frac{n}{5^i}\right) + i \cdot n^2$$

Caso base

$$\frac{n}{5^1} = 1 \rightarrow n = 5^1 \rightarrow \log_5(n) = i$$

Reemplazo el valor de i en el paso general

$$5^{i \cdot 2} T\left(\frac{n}{5^i}\right) + i \cdot n^2 = 5^{\log_5(n) \cdot 2} T\left(\frac{n}{5^{\log_5(n)}}\right) + \log_5(n) \cdot n^2 =$$

$$= 5^{\log_5(n) \cdot 2} \cdot T(1) + \log_5(n) \cdot n^2 = n \cdot 2 \cdot T(1) + \log_5(n) \cdot n^2 =$$

$$= n^2 \cdot 1 + n^2 \cdot \log_5(n) = n^2 + n^2 \cdot \log_5(n)$$

$$T(n) = n^2 + n^2 \cdot \log_5(n) = O(n^2 \cdot \log_5(n))$$